Unit IV: Sampling Methods

Table of Contents

1. Introduction to Sampling
2. Approximate Inference
3. Monte Carlo Sampling Methods
4. Sampling Methods for Pattern Recognition
5. Continuous Latent Variables
6. Advanced Sampling Techniques
7. Real-World Applications
8. Key Formulas Summary

1. Introduction to Sampling

What is Sampling?

Simple Definition: Sampling is like taking a small representative portion from a large population to understand the whole group.

In Pattern Recognition: We use sampling to:

- Make predictions when exact calculations are too difficult
- Handle large datasets efficiently
- Deal with uncertainty in our models

Why Do We Need Sampling?

Point-by-Point Reasons:

- **Computational Limits**: Some mathematical calculations are impossible to solve exactly
- **Time Constraints**: Exact methods take too long for real-world applications
- **Memory Issues**: Large datasets don't fit in computer memory
- **Practical Solutions**: Sampling gives "good enough" answers quickly

Types of Sampling Methods

1. Direct Sampling

- **What it is**: Taking samples directly from known distributions
- **Example**: Rolling a dice to get random numbers 1-6
- **When to use**: When we know the exact distribution

2. Rejection Sampling

- **What it is**: Generate samples and accept/reject based on criteria
- **Example**: Throwing darts at a board and keeping only those that hit the target
- **When to use**: When direct sampling is difficult

3. Importance Sampling

- **What it is**: Sample from an easier distribution and adjust with weights
- **Example**: Survey more people in cities (easier) but weight rural responses higher
- **When to use**: When target distribution is hard to sample from

4. MCMC (Markov Chain Monte Carlo)

- **What it is**: Generate sequences of related samples
- **Example**: Random walk where next step depends on current position
- **When to use**: For complex, high-dimensional problems

Applications in Pattern Recognition

Common Uses:

- **Image Recognition**: Identifying objects in photos
- **Speech Recognition**: Converting speech to text
- **Medical Diagnosis**: Analyzing symptoms to predict diseases
- **Recommendation Systems**: Suggesting products or content
- **Financial Modeling**: Predicting market trends

2. Approximate Inference

Need for Approximate Inference

What is Inference?

- **Simple Definition**: Making conclusions based on available information
- **In Pattern Recognition**: Using data to make predictions or decisions

Why Approximate?

- **Exact inference** is often impossible due to:
  - Too many variables to consider
  - Complex mathematical relationships
  - Limited computational resources
  - Time constraints in real applications

Real-World Example: Weather prediction uses approximate inference because it's impossible to measure every air molecule, but we can still make useful forecasts.

Exact vs Approximate Inference

Exact Inference:

- **Advantages**: Perfect accuracy, no errors
- **Disadvantages**: Often impossible, very slow, limited to simple problems
- **Example**: Calculating exact probability of getting heads in a coin flip (50%)

Approximate Inference:

- **Advantages**: Practical, fast, works for complex problems
- **Disadvantages**: Some error, results may vary slightly
- **Example**: Estimating probability by flipping coin 100 times (might get 48% or 52%)

Deterministic vs Stochastic Inference

Deterministic Methods:

- **What it means**: Same input always gives same output
- **Characteristics**: Predictable, repeatable results
- **Example**: Using a formula to calculate area of a circle
- **Methods**: Variational inference, Laplace approximation

Stochastic Methods:

- **What it means**: Involves randomness, results may vary slightly
- **Characteristics**: Uses random sampling, different runs give slightly different results
- **Example**: Monte Carlo simulation
- **Methods**: MCMC, particle filtering

Approximation Accuracy vs Computational Cost

The Trade-off:

- **More accuracy** = More computation time and resources
- **Less accuracy** = Faster results, less resources needed

Factors to Consider:

- **Sample Size**: More samples = better accuracy but slower
- **Algorithm Complexity**: Sophisticated methods = better results but more complex
- **Problem Difficulty**: Harder problems need more computation
- **Time Constraints**: Real-time applications need fast approximate solutions

Practical Example: GPS navigation uses approximate methods to quickly find routes rather than checking every possible path (which would take hours).

Role in Probabilistic Models

What are Probabilistic Models?

- Models that deal with uncertainty and probability
- Help make decisions when we're not 100% sure about outcomes

Key Functions:

- **Parameter Estimation**: Learning the best values for model settings
- **Prediction**: Making forecasts about future events
- **Model Selection**: Choosing the best model among alternatives
- **Uncertainty Quantification**: Measuring how confident we are in our predictions

Example: Email spam detection uses probabilistic models to estimate the probability that an email is spam based on words and patterns.

3. Monte Carlo Sampling Methods

Monte Carlo Sampling Intuition

What is Monte Carlo?

- **Named after**: Monte Carlo casino (because it involves randomness like gambling)
- **Basic Idea**: Use random sampling to solve mathematical problems
- **Key Principle**: If you can't solve something exactly, try it many times randomly and average the results

Simple Example: To estimate the area of an irregular shape, throw random dots at it and count how many land inside vs outside.

Why It Works:

- **Law of Large Numbers**: As you increase samples, your estimate gets closer to the true answer
- **Universal Method**: Works for any problem you can simulate
- **Scalable**: Can handle problems with many variables

Random Sampling from Distributions

What is a Distribution?

- **Simple Definition**: A description of how likely different outcomes are
- **Example**: Height distribution shows most people are average height, few are very tall or very short

Common Distributions:

1. Uniform Distribution

- **What it is**: All outcomes equally likely
- **Example**: Rolling a fair dice (each number 1-6 has equal chance)
- **Formula**: All values between a and b have equal probability

2. Normal (Gaussian) Distribution

- **What it is**: Bell-shaped curve, most values near the center
- **Example**: Human heights, test scores
- **Formula**: $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

3. Exponential Distribution

- **What it is**: Models waiting times
- **Example**: Time between phone calls, equipment failures
- **Formula**: $f(x) = \lambda e^{-\lambda x}$

Importance of Sampling in Posterior Estimation

What is Posterior?

- **Simple Definition**: Updated belief after seeing new evidence
- **Example**: Your belief about weather after looking outside

Bayes' Rule (Simple Form):

$$\text{Posterior} = \frac{\text{Likelihood} \times \text{Prior}}{\text{Evidence}}$$

Why Sampling is Important:

- **Direct Calculation**: Often impossible for complex problems
- **Sampling Solution**: Generate many samples from posterior distribution
- **Practical Benefit**: Can estimate any quantity of interest from samples

Example: Medical diagnosis updates disease probability after test results using posterior estimation.

Sampling-based Expectation Estimation

What is Expectation?

- **Simple Definition**: Average value you expect to get
- **Mathematical Definition**: Weighted average of all possible outcomes

Basic Monte Carlo Estimation:

$$\text{Average} \approx \frac{1}{N}\sum_{i=1}^{N} f\,(x_i)$$

Steps:

1. Generate N random samples
2. Apply function f to each sample
3. Take the average of all results

Example: To estimate average height in a city, randomly sample 1000 people and calculate their average height.

Convergence and Variance Considerations

What is Convergence?

- **Simple Definition**: Getting closer to the true answer as you use more samples
- **Key Point**: More samples = better estimate (usually)

Convergence Rate:

- **Rule of Thumb**: Error decreases as $\frac{1}{\sqrt{N}}$ where N is number of samples
- **Practical Meaning**: To halve the error, you need 4 times more samples

What is Variance?

- **Simple Definition**: How much your estimates vary between different runs
- **High Variance**: Results change a lot between runs
- **Low Variance**: Results are consistent

Variance Reduction Techniques:

- **Control Variates**: Use known information to reduce uncertainty
- **Stratified Sampling**: Divide problem into parts and sample each part
- **Importance Sampling**: Focus sampling on important regions

4. Sampling Methods for Pattern Recognition

Monte Carlo Sampling Intuition in Pattern Recognition

Pattern Recognition Tasks:

- **Classification**: Deciding which category something belongs to
- **Clustering**: Grouping similar items together
- **Regression**: Predicting numerical values

- **Dimensionality Reduction**: Simplifying complex data

Why Use Sampling?

- **High-Dimensional Data**: Images, text, and audio have thousands of features
- **Complex Models**: Neural networks and Bayesian models are too complex for exact solutions
- **Real-Time Requirements**: Need fast approximate answers

Example: Face recognition in photos uses sampling to handle millions of pixel values efficiently.

Random Sampling from Distributions

In Pattern Recognition Context:

1. Data Sampling

- **Bootstrap Sampling**: Randomly sample data with replacement
- **Cross-Validation**: Randomly split data into training and testing sets
- **Example**: Training a model on random subsets of data to improve generalization

2. Parameter Sampling

- **Bayesian Neural Networks**: Sample different network weights
- **Ensemble Methods**: Create multiple models with random parameters
- **Example**: Random Forest uses random subsets of features for each tree

3. Feature Sampling

- **Random Projections**: Randomly combine features to reduce dimensions
- **Dropout**: Randomly ignore some neurons during training
- **Example**: Randomly selecting 100 features from 10,000 available features

Importance of Sampling in Posterior Estimation

Posterior in Pattern Recognition:

- **What it represents**: Probability of model parameters given observed data
- **Why it matters**: Tells us how confident we should be in our model

Applications:

1. Bayesian Classification

- **Goal**: Estimate probability that an item belongs to each class
- **Method**: Sample from posterior distribution of class probabilities
- **Example**: Email spam detection with confidence scores

2. Parameter Uncertainty

- **Goal**: Understand how uncertain we are about model parameters
- **Method**: Sample different parameter values from posterior
- **Example**: Medical diagnosis with uncertainty bounds

3. Model Selection

- **Goal**: Choose the best model among alternatives
- **Method**: Compare posterior probabilities of different models
- **Example**: Choosing between linear and nonlinear models

Sampling-based Expectation Estimation

Expectation in Pattern Recognition:

1. Prediction

- **Goal**: Estimate expected output for new inputs
- **Method**: Average predictions from multiple sampled models
- **Formula**: $E[y|x] \approx \frac{1}{N}\sum_{i=1}^{N} f\,(x, \theta_i)$

2. Risk Assessment

- **Goal**: Estimate expected loss or error
- **Method**: Sample different scenarios and compute average loss
- **Example**: Estimating classification error rate

3. Feature Importance

- **Goal**: Determine which features are most important
- **Method**: Sample different feature subsets and measure performance
- **Example**: Gene selection in medical diagnosis

Convergence and Variance Considerations

Convergence in Pattern Recognition:

1. Training Convergence

- **What to monitor**: Loss function, accuracy metrics
- **Signs of convergence**: Metrics stop improving significantly
- **Example**: Neural network training loss plateaus

2. Sampling Convergence

- **What to monitor**: Sample statistics, trace plots
- **Signs of convergence**: Sample means stabilize

- **Example**: MCMC chains reach stationary distribution

Variance Considerations:

1. Bias-Variance Trade-off

- **High Bias**: Model is too simple, underfits data
- **High Variance**: Model is too complex, overfits data
- **Goal**: Find balance between bias and variance

2. Ensemble Variance

- **Single Model**: High variance in predictions
- **Multiple Models**: Lower variance through averaging
- **Example**: Random Forest reduces variance compared to single decision tree

5. Continuous Latent Variables

Latent Variable Concept

What are Latent Variables?

- **Simple Definition**: Hidden factors that influence what we observe but can't be measured directly
- **Analogy**: Like the "mood" of a person - you can't measure it directly, but it affects their behavior
- **Other Names**: Hidden variables, unobserved variables, factors

Examples in Daily Life:

- **Intelligence**: Can't measure directly, but affects test scores, problem-solving ability
- **Customer Satisfaction**: Hidden, but influences purchase behavior, reviews
- **Disease Severity**: Not directly observable, but affects symptoms

Mathematical Representation:

- **Observed Data**: X (what we can see/measure)
- **Latent Variables**: Z (hidden factors)
- **Relationship**: X depends on Z, but we only observe X

Continuous vs Discrete Latent Variables

Discrete Latent Variables:

- **What they are**: Hidden categories or groups
- **Examples**:
  - Customer types (budget-conscious, luxury-seeking, convenience-focused)
  - Disease types (Type A, Type B, Type C)

  - Topic categories in documents (sports, politics, entertainment)
- **Mathematical Form**: $Z \in \{1, 2, 3, ..., K\}$

Continuous Latent Variables:

- **What they are**: Hidden factors that can take any value in a range
- **Examples**:
  - Skill level (can be any value from 0 to 100)
  - Mood intensity (from very sad to very happy)
  - Economic factors (inflation rate, growth rate)
- **Mathematical Form**: $Z \in \mathbb{R}$ (real numbers)

Comparison Table:

Aspect	Discrete	Continuous
Values	Categories (1,2,3...)	Any real number
Examples	Customer types	Skill levels
Math Operations	Summation	Integration
Complexity	Simpler	More complex

Generative Models with Latent Variables

What are Generative Models?

- **Simple Definition**: Models that can create new data similar to what they've seen
- **Process**: Learn hidden patterns, then generate new examples
- **Example**: A model trained on photos can generate new realistic photos

How Latent Variables Help:

- **Compression**: Represent complex data with simpler hidden factors
- **Generation**: Create new data by sampling from latent space
- **Understanding**: Discover meaningful hidden patterns

Common Generative Models:

1. Variational Autoencoder (VAE)

- **Purpose**: Learn compressed representations of data
- **Process**:
  - Encoder: Data → Latent variables
  - Decoder: Latent variables → Reconstructed data
- **Example**: Compress images into small codes, then reconstruct them

2. Gaussian Mixture Model (GMM)

- **Purpose**: Model data as mixture of groups
- **Latent Variable**: Which group each data point belongs to

- **Example**: Customer segmentation based on purchase behavior

3. Factor Analysis

- **Purpose**: Find hidden factors that explain correlations in data
- **Process**: Express observed variables as combinations of hidden factors
- **Example**: Intelligence factors explaining performance on different tests

Marginalization over Latent Variables

What is Marginalization?

- **Simple Definition**: Removing/integrating out variables we don't care about
- **Analogy**: Like getting total sales by adding up sales from all regions (ignoring regional breakdown)
- **Purpose**: Focus on what matters, ignore irrelevant details

Why is it Important?

- **Prediction**: We often care about outputs, not hidden variables
- **Simplification**: Reduces complexity by removing unnecessary variables
- **Inference**: Allows us to make conclusions about observed data

Mathematical Process: For discrete latent variables: Sum over all possible values For continuous latent variables: Integrate over all possible values

Example: In customer analysis, we might care about total purchase amount (marginalize over specific product categories).

Challenges:

- **Computational Difficulty**: Integration/summation can be very complex
- **High Dimensions**: Many latent variables make computation harder
- **Unknown Distributions**: Don't know exact form of latent variable distributions

Solutions:

- **Sampling Methods**: Use Monte Carlo to approximate integrals
- **Variational Methods**: Find simpler approximating distributions
- **EM Algorithm**: Iteratively estimate latent variables and parameters

Role in Clustering and Density Estimation

Clustering with Latent Variables:

What is Clustering?

- **Simple Definition**: Grouping similar items together
- **Latent Variable Role**: Hidden group membership for each data point
- **Example**: Grouping customers by shopping behavior (but don't know groups beforehand)

Gaussian Mixture Model for Clustering:

- **Idea**: Data comes from multiple Gaussian distributions (clusters)
- **Latent Variable**: Which cluster each point belongs to
- **Process**:
    1. Assume K clusters exist
    2. Estimate cluster parameters (mean, variance)
    3. Assign points to most likely cluster

Density Estimation with Latent Variables:

What is Density Estimation?

- **Simple Definition**: Estimating how likely different values are in your data
- **Purpose**: Understand data distribution, detect outliers, generate new samples

How Latent Variables Help:

- **Flexibility**: Can model complex, multi-modal distributions
- **Interpretability**: Latent variables often have meaningful interpretations
- **Generalization**: Better performance on new, unseen data

Example Applications:

- **Anomaly Detection**: Points with low density are outliers
- **Data Generation**: Sample from estimated density to create new data
- **Compression**: Use latent variables as compressed representation

Practical Benefits:

- **Automatic Discovery**: Find hidden patterns without manual specification
- **Dimensionality Reduction**: Represent high-dimensional data with fewer latent variables
- **Uncertainty Quantification**: Measure confidence in cluster assignments

6. Advanced Sampling Techniques

Markov Chain Monte Carlo (MCMC)

What is MCMC?

- **Simple Definition**: A method that creates a sequence of samples where each sample depends on the previous one
- **Analogy**: Like a random walk where your next step depends on where you are now
- **Key Idea**: Eventually, this walk explores the entire distribution we're interested in

Why Use MCMC?

- **Complex Distributions**: When direct sampling is impossible

- **High Dimensions**: Works well even with many variables
- **Flexibility**: Can handle almost any probability distribution

Basic MCMC Process:

1. Start at some initial point
2. Propose a new point based on current location
3. Decide whether to accept or reject the new point
4. Repeat many times
5. Use the sequence of accepted points as samples

Metropolis-Hastings Algorithm:

- **Most Common MCMC Method**
- **Steps**:
    1. Propose new state based on current state
    2. Calculate acceptance probability
    3. Accept with calculated probability, otherwise stay at current state
- **Acceptance Rule**: Accept if new state is better, sometimes accept even if worse

Gibbs Sampling

What is Gibbs Sampling?

- **Simple Definition**: Sample each variable one at a time, keeping others fixed
- **Analogy**: Like adjusting one knob at a time on a radio to find the best station
- **Special Case**: Type of MCMC that's often easier to implement

When to Use Gibbs Sampling:

- **Conditional Distributions**: When it's easy to sample each variable given the others
- **High Dimensions**: Breaks complex problem into simpler pieces
- **Conjugate Models**: Works especially well with certain mathematical structures

Gibbs Sampling Process:

1. Initialize all variables
2. For each variable in turn:
    - Sample new value given current values of all other variables
    - Update that variable
3. Repeat until convergence

Example: In image denoising, sample each pixel's true value given its neighbors and the noisy observation.

Hamiltonian Monte Carlo (HMC)

What is HMC?

- **Simple Definition**: Uses physics concepts to make MCMC more efficient
- **Physics Analogy**: Like a ball rolling on a landscape - it naturally moves toward good regions
- **Key Advantage**: Takes larger, more informed steps than random walk

Why HMC is Better:

- **Faster Convergence**: Reaches target distribution more quickly
- **Less Correlation**: Successive samples are more independent
- **Better Exploration**: Avoids getting stuck in local regions

Basic Concept:

- **Potential Energy**: Related to probability (high probability = low energy)
- **Kinetic Energy**: Momentum that helps exploration
- **Dynamics**: Use physics equations to propose good moves

When to Use HMC:

- **Smooth Distributions**: Works best when you can compute gradients
- **High Dimensions**: Particularly effective in many dimensions
- **Complex Posteriors**: Good for multimodal or correlated distributions

Particle Filtering (Sequential Monte Carlo)

What is Particle Filtering?

- **Simple Definition**: Track multiple possible states (particles) over time
- **Analogy**: Like having multiple hypotheses about where a moving object is
- **Key Idea**: Update beliefs as new observations arrive

Why Use Particle Filtering?

- **Dynamic Systems**: Things that change over time
- **Nonlinear Models**: When simple methods like Kalman filters don't work
- **Non-Gaussian Noise**: Can handle any type of uncertainty

Particle Filter Process:

1. **Prediction**: Move each particle according to system dynamics
2. **Update**: Weight particles based on how well they match observations
3. **Resampling**: Keep particles with high weights, discard low-weight particles
4. **Repeat**: Continue for next time step

Applications:

- **Object Tracking**: Following moving objects in video
- **Robot Localization**: Determining robot position from sensor data
- **Financial Modeling**: Tracking market states over time

Example: Tracking a person walking through a building using noisy GPS signals - maintain multiple hypotheses about their location.

Variational Inference

What is Variational Inference?

- **Simple Definition**: Turn sampling problem into optimization problem
- **Key Idea**: Find the best simple approximation to a complex distribution
- **Advantage**: Often faster than sampling methods

Why Use Variational Inference?

- **Speed**: Optimization is often faster than sampling
- **Deterministic**: Same input always gives same output
- **Scalable**: Works well with large datasets

Basic Approach:

1. Choose a family of simple distributions (e.g., Gaussian)
2. Find the member of this family that best approximates the target
3. "Best" means minimizing some measure of difference

Trade-offs:

- **Pros**: Fast, deterministic, scalable
- **Cons**: Limited by choice of approximating family, may miss important features

Applications:

- **Large-Scale Bayesian Models**: When MCMC is too slow
- **Deep Learning**: Bayesian neural networks
- **Topic Modeling**: Large document collections

7. Key Formulas Summary

Basic Probability

Bayes' Rule:

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

Expectation:

$$E[X] = \sum_{x} x \times P(X = x)$$

(discrete)

$$E[X] = \int x \times f(x)dx$$

(continuous)

Monte Carlo Estimation

Basic Estimator:

$$\hat{\mu} = \frac{1}{N}\sum_{i=1}^{N} f\,(x_i)$$

Standard Error:

$$SE = \frac{\sigma}{\sqrt{N}}$$

Importance Sampling

Weighted Estimator:

$$\hat{\mu} = \frac{1}{N}\sum_{i=1}^{N} f\,(x_i) \times w_i$$

Importance Weights:

$$w_i = \frac{p(x_i)}{q(x_i)}$$

MCMC

Metropolis-Hastings Acceptance:

$$\alpha = \min\left(1, \frac{p(\theta_{new})}{p(\theta_{old})}\right)$$

Gaussian Distribution

Univariate:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Multivariate:

$$f(x) = \frac{1}{(2\pi)^{k/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right)$$

Summary

Key Takeaways

1. Sampling is Essential

- Many real-world problems can't be solved exactly
- Sampling provides practical approximate solutions
- Trade-off between accuracy and computational cost

2. Different Methods for Different Problems

- **Direct sampling**: When you know the distribution
- **MCMC**: For complex, high-dimensional problems
- **Importance sampling**: When target distribution is hard to sample
- **Particle filtering**: For dynamic, time-varying systems

3. Latent Variables are Powerful

- Represent hidden factors that explain observed data
- Enable discovery of meaningful patterns
- Essential for clustering and generative modeling

4. Real-World Impact

- Image recognition and computer vision
- Natural language processing and topic modeling
- Financial risk assessment and decision making
- Scientific modeling and data analysis

5. Practical Considerations

- Always check convergence of sampling algorithms
- Consider computational resources and time constraints
- Validate results with multiple methods when possible
- Understand limitations and assumptions of each method

Problems:

Monte Carlo:

A factory wants to estimate the **expected time (in minutes)** needed to complete a small machine task.
They assume the task time varies uniformly between **3 and 8 minutes**.
They record the following 5 observed task times:

$$4,\ 7,\ 5,\ 6,\ 8$$

Use the Monte Carlo method to estimate the expected task time.

Solution:

1. Write the Monte Carlo estimator

$$\widehat{\mathbb{E}}[X] = \frac{1}{N}\sum_{i=1}^{N} X_i$$

Where

- $N = 5$
- $X_i$are the observed task times

2. Substitute the values

$$\widehat{\mathbb{E}}[X] = \frac{4+7+5+6+8}{5}$$

3. Compute the sum

$$4+7+5+6+8 = 30$$

4. Divide by the number of observations

$$\widehat{\mathbb{E}}[X] = \frac{30}{5} = 6$$

Final Answer

$\boxed{\text{6 minutes}}$

KL Divergence:

A small online bookstore tracks the types of books its customers purchase.
Based on last month's data, the store estimates that the probability distribution of purchases across three categories is:

$$P = \{\text{Fiction: } 0.5,\ \text{Science: } 0.3,\ \text{History: } 0.2\}$$

This month, the marketing team predicts a slightly different distribution:

$$Q = \{\text{Fiction: } 0.4,\ \text{Science: } 0.4,\ \text{History: } 0.2\}$$

Using **KL divergence**, compute how different last month's actual distribution $P$is from this month's predicted distribution Q.

Solution:
KL divergence for discrete distributions is:

$$D_{KL}(P\ ||\ Q) = \sum_i P(i) \log\left(\frac{P(i)}{Q(i)}\right)$$

1. Write each term

Fiction:

$$0.5\log\left(\frac{0.5}{0.4}\right)$$

Science:

$$0.3\log\left(\frac{0.3}{0.4}\right)$$

History:

$$0.2\log\left(\frac{0.2}{0.2}\right)$$

2. Compute the ratios

$$\frac{0.5}{0.4} = 1.25$$

$$\frac{0.3}{0.4} = 0.75$$

$$\frac{0.2}{0.2} = 1$$

3. Apply natural log

$$\log(1.25) = 0.2231$$

$$\log(0.75) = -0.2877$$

$$\log(1) = 0$$

4. Multiply by the probabilities

$$0.5 \times 0.2231 = 0.11155$$

$$0.3 \times (-0.2877) = -0.08631$$

$$0.2 \times 0 = 0$$

5. Add all contributions

$$D_{KL}(P \mid\mid Q) = 0.11155 - 0.08631 + 0$$

$$D_{KL}(P \mid\mid Q) = 0.02524$$

Final Answer

$$\boxed{D_{KL}(P \mid\mid Q) \approx 0.025}$$

The KL value is small, meaning the predicted distribution is fairly close to the actual distribution.

Sampling Variance:

A company wants to estimate how much the daily electricity usage of its office fluctuates. They model the daily usage X(in units) as a random variable and record the following 5 sampled observations from different days:

$$12,\ 15,\ 14,\ 10,\ 13$$

Using **sampling methods**, compute the **sample variance** of the electricity usage based on these values.

Solution:

1. Write the sample variance formula

For Nsamples, sample variance is:

$$s^2 = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2$$

2. Compute the sample mean

$$\bar{x} = \frac{12 + 15 + 14 + 10 + 13}{5}$$

$$\bar{x} = \frac{64}{5} = 12.8$$

3. Compute each squared deviation

$$(12 - 12.8)^2 = 0.64$$

$$(15 - 12.8)^2 = 4.84$$

$$(14 - 12.8)^2 = 1.44$$

$$(10 - 12.8)^2 = 7.84$$

$$(13 - 12.8)^2 = 0.04$$

4. Sum the squared deviations

$$0.64 + 4.84 + 1.44 + 7.84 + 0.04 = 14.8$$

5. Divide by $N - 1 = 4$

$$s^2 = \frac{14.8}{4} = 3.7$$

Final Answer

$$\boxed{s^2 = 3.7}$$

The sampling variance indicates a moderate fluctuation in daily electricity usage.